

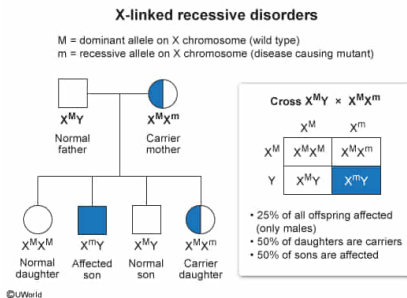
A woman carries an X-linked recessive allele for a disorder characterized by muscle degeneration but does not express this condition herself. If she has multiple offspring with an unaffected man, what proportion of the woman's children will be expected to exhibit this degenerative disorder?

- ☐ A. All
- ☐ B. Half [14%]
- ☐ C. None [19%]
- ✓ ☒ D. One quarter [65%]

Correct

66% answered correctly

Explanation:



Sex linkage is the expression of an allele located on a sex chromosome (eg, X or Y in humans). In **X-linked recessive disorders**, females (XX) express the recessive phenotype only when *both* X chromosomes (inherited from their mother and father) have the recessive allele. In contrast, males (XY) need only inherit a single copy of the recessive allele from their mother to exhibit the disease.

The given scenario describes the genetic transmission of an X-linked recessive disorder characterized by muscle degeneration. The mother is a carrier, meaning she has the disease-causing allele but does not express the disorder, whereas the father does not carry the allele and is unaffected by the disorder. Genetically, the woman has one copy of the wild-type dominant allele on one X chromosome (X^M) and one copy of the recessive allele on the other X chromosome (X^m). The woman will pass one of her two X chromosomes to each child; therefore, each child has a **50% chance of receiving the recessive allele**, regardless of gender. In contrast, the man will pass his dominant X chromosome to all female offspring (50% chance) and his Y chromosome to male offspring (**50% chance**).

Accordingly, none of their daughters are expected to be affected but will be either homozygous dominant ($X^M X^M$) or carriers of the mutation ($X^M X^m$). However, the sons will inherit a Y chromosome and either the dominant allele (and be unaffected) or the recessive allele (and express the disorder). Consequently, the **overall probability (P)** that a child will express an X-linked recessive trait can be calculated as:

$$P(\text{child inherits affected X-chromosome from mother}) \times P(\text{child is male}) = 0.50 \times 0.50 = 0.25$$

Therefore, **one quarter (25%)** of the offspring are *expected* to express the disorder characterized by muscle degeneration (note that the *expected* proportion does not mean 25% of their children *will* exhibit the disorder, only that this is the expected proportion assuming all alleles are distributed evenly).

(Choice B) Half (50%) of the children will inherit the recessive allele. However, only male children will express the disorder.

Educational objective:

For the cross between a female carrier of an X-linked recessive trait and an unaffected man, only male offspring will express the trait. The probability of a child having the trait is given by $P(\text{child inherits affected X-chromosome from mother}) \times P(\text{child is male}) = 0.50 \times 0.50 = 0.25$. Therefore, 25% of all offspring are expected to express the trait.

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